



ISLAMIC UNIVERSITY OF TECHNOLOGY

Electronics Mini Project

Team UNBIASED Transistor

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Implementation of Demultiplexer System System Using BJTs

Description:

Designed and implemented a **1-to-2 demultiplexer** using discrete **BJTs**. The circuit routes a single digital input to one of two outputs based on the address (select) signal. The outputs are implemented as **active-low transistor switches**, driving LEDs to indicate the selected output.

The project demonstrates how digital logic functions can be realized using discrete transistor circuits instead of integrated logic ICs.

Using the design of **1-to-2 demultiplexer** we can build even more intricate systems like **1-to-4** or even **1-to-8** demultiplexers.

Proposed System Overview:

The proposed system consists of **three main sections**:

Input Section:

- Accepts a single digital input signal.
- Receives one or more address (select) inputs.
- Provides the control signals required for signal routing.

Demultiplexer Logic Section:

- Implements demultiplexer logic using discrete **BC547 NPN BJTs**.
- Uses transistors as electronic switches to realize digital logic.
- Routes the input signal to the selected output based on the address input.
- Can be configured as a **1-to-2** or expanded to a **1-to-4** demultiplexer.

Output Section:

- Produces one active output corresponding to the selected address.
- Uses LEDs as output indicators.
- Demonstrates successful signal routing through transistor switching

Working Principle:

When the input signal is LOW (0):

- All output LEDs remain OFF.
- No output channel is activated.

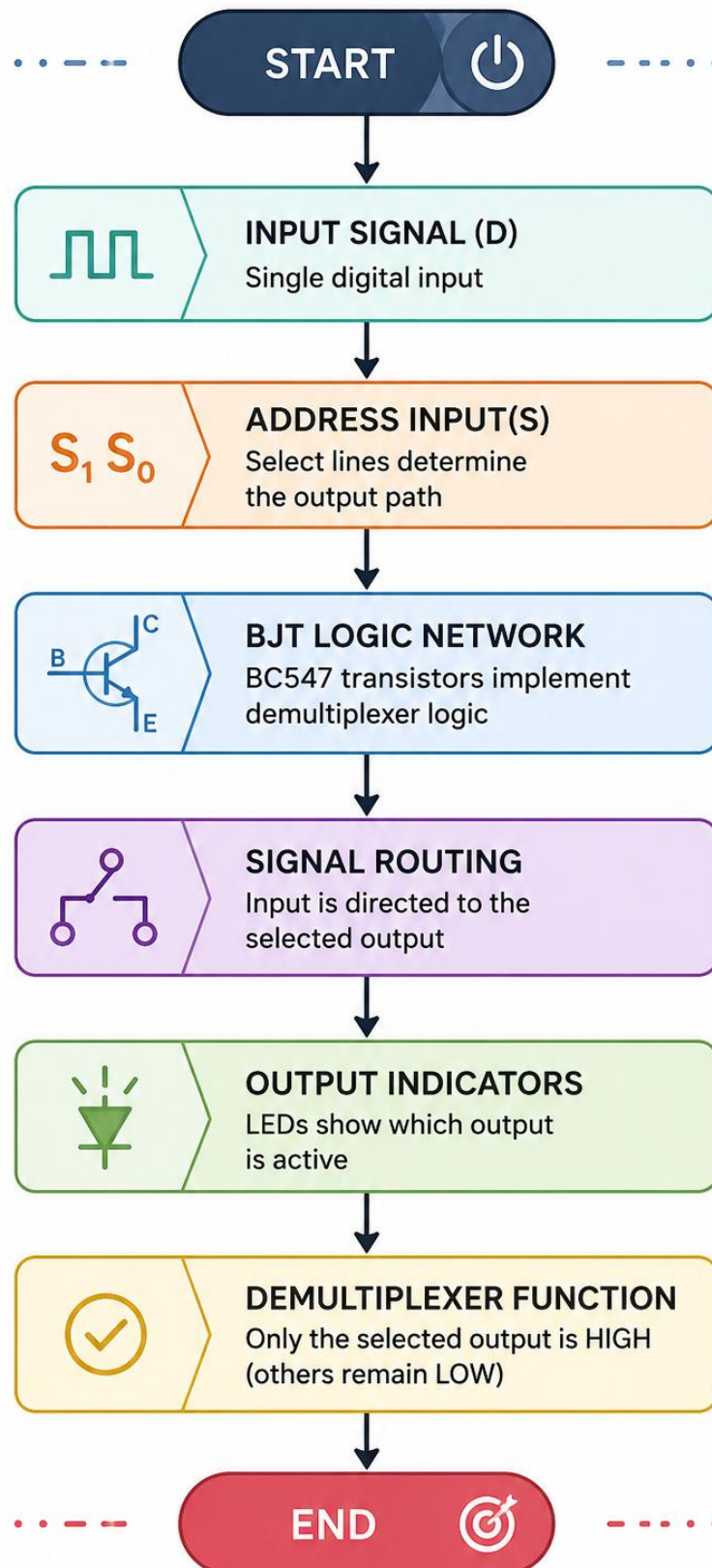
When the input signal is HIGH (1):

- The address (select) input determines the active output.
- The selected transistor path turns ON.
- Only the corresponding output LED lights up.
- All other outputs remain inactive.

Use of BJT:

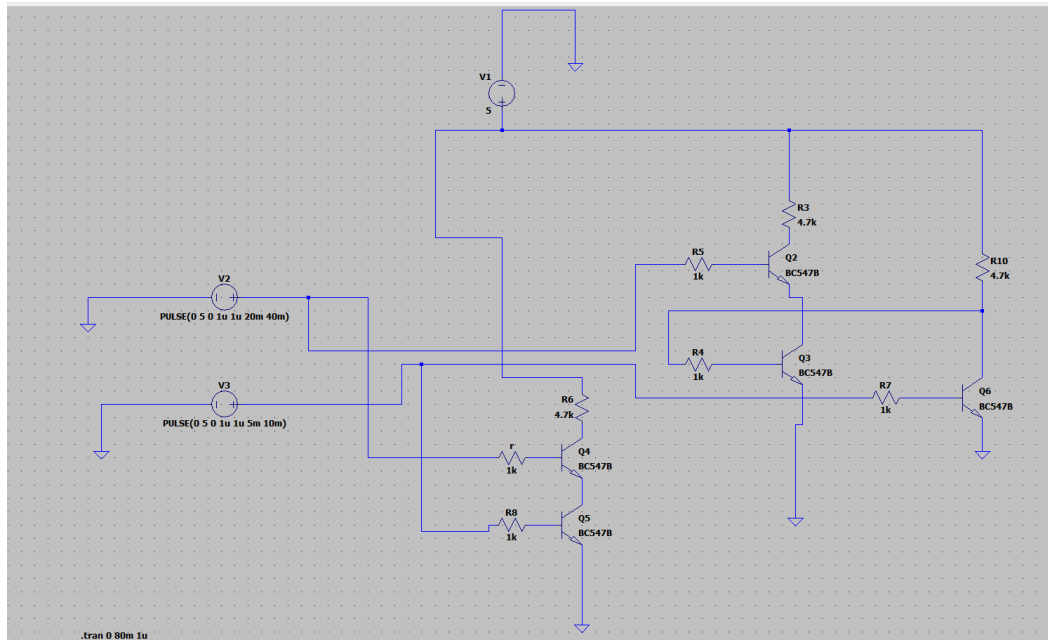
1. **Acts as the primary switching element** for implementing signal routing network.
2. **Implements discrete logic functions** required for the routing mechanism
3. **Controls the selection of one output channel** based on the applied input and control signals
4. **Demonstrates the application of BJTs in digital circuit design** using discrete components instead of dedicated logic ICs.
5. Some BJTs provide **logic inversion** required to realize the demultiplexer operation.
6. The **output BJTs sink current** to drive the LEDs **safely**.

FlowChart:

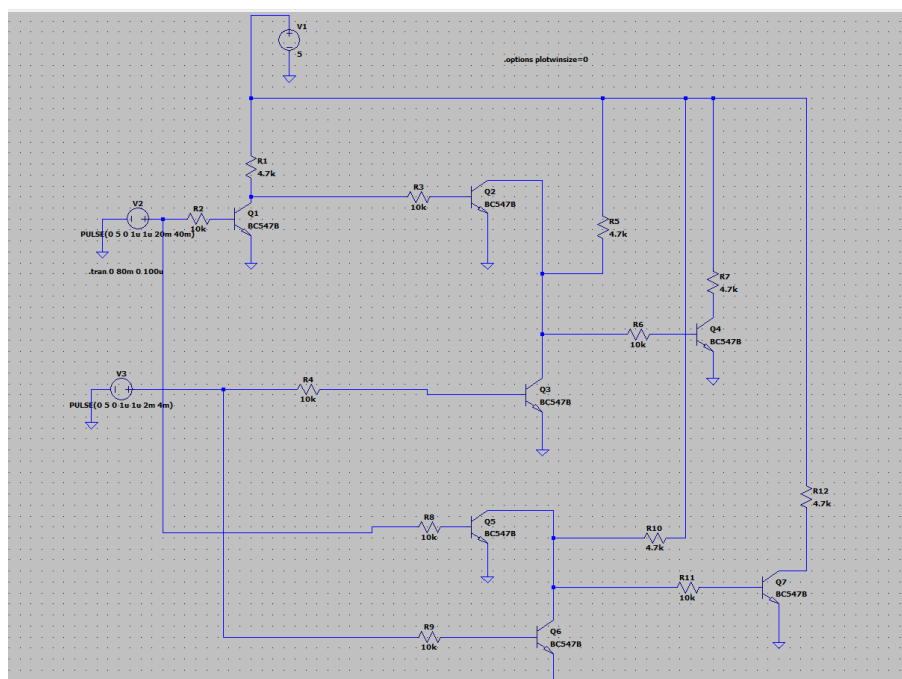


Circuit Diagram:

We will be using either or both of these two Circuit Diagrams to implement our Demultiplexer Circuit:



Circuit – 1



Circuit – 2

Components:

Sl. No.	Component	Specification	Purpose
1	BC547 NPN Transistor	BC547B	Implements demultiplexer logic and switching
2	Resistors	4.7 k Ω , 10 k Ω	Current limiting and transistor biasing
3	LEDs	5 mm Red/Green	Output indication
4	DC Power Supply	+5 V	Powers the circuit
5	Breadboard	Standard	Circuit prototyping
6	Jumper Wires	Male-to-Male	Electrical connections
7	Function Generator / Pulse Source	0–5 V Digital Pulse	Generates input and address signals
8	Multimeter (Optional)	Digital	Voltage and continuity testing
9	Oscilloscope (Optional)	Digital	Signal waveform observation

Engineering Design Considerations:

- ***Use of Discrete BJTs:*** The demultiplexer logic is implemented using **BC547 NPN** transistors instead of digital logic ICs to demonstrate transistor-level digital circuit design.
- ***Supply Voltage:*** The circuit operates from a **+5 V DC supply**, making it compatible with standard digital logic levels and suitable for laboratory implementation.
- ***Transistor Switching:*** BC547 transistors are operated in **cut-off and saturation regions** to function as electronic switches for digital signal routing.
- ***Resistor Selection:*** Appropriate **4.7 k Ω** and **10 k Ω** resistors are used to provide transistor biasing, limit base current, and ensure reliable switching.
- ***Modular Design:*** The circuit is designed with a modular structure so that it can be expanded from a **1-to-2 demultiplexer** to a **1-to-4 or larger configuration** if required.
- ***Output Indication:*** LEDs are connected to the outputs to provide a clear visual indication of the selected output channel during testing and demonstration.

- ***Simulation Before Hardware:*** The circuit is first verified using simulation software before physical implementation to identify design errors and improve reliability.
- ***Scalability:*** The design principle allows additional output channels to be added by extending the transistor logic network.
- ***Low-Cost Implementation:*** The circuit uses inexpensive and readily available components, making it economical and easy to reproduce for educational purposes.
- ***Ease of Troubleshooting:*** The circuit is divided into input, logic, and output stages, simplifying testing, debugging, and future modifications.

Expected Outcome:

- Successful implementation of a **transistor-based demultiplexer** using BC547 NPN BJTs.
- Correct routing of a single digital input signal to the selected output based on the address (select) input(s).
- Accurate demonstration of digital switching using discrete transistor circuits.

- LEDs provide clear visual indication of the active output channel.
- Stable operation with a **+5 V DC** power supply.
- Verification of the circuit's functionality through simulation and hardware testing.
- A modular design that can be expanded from a basic configuration to support additional output channels (e.g., from **1-to-2** to **1-to-4**).
- Improved understanding of BJT switching, digital logic implementation, and signal routing using discrete electronic components.